



Illustration: Pradeep Kumar

who is responsible for it. If the product fails, each team member is responsible for it.

Peer review is an important part of team work. In this process, a team member requests an artefact to be reviewed. The other team members then provide their review comments, which may include corrections, suggestions and doubts about the artefact. The artefact is then updated based on the review comments. This process is repeated till the artefact is up to the satisfaction of all the team members. Of course, in case of any conflict, it is the project leader who makes the final decision. Figure 2 gives a pictorial representation of the procedure.

Reviews are critical at the following stages of software development:

1. Requirement-specification review: As discussed earlier, understanding the customers' requirements is very important. So, while you prepare the requirement specification, get it reviewed by your team members. Ten people in a team may have

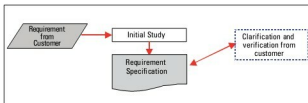


Figure 1: Analysing application users' requirements

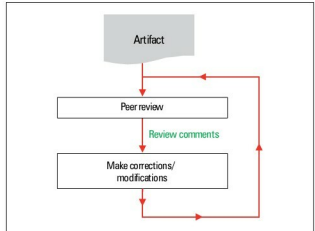


Figure 2: Steps involved in peer review

ten different interpretations of the specifications. Discuss it within your team, and then pass on your understanding to the customer for verification.

2. Design review: Once the requirement specification gets finalised, we move on to the design phase. In the design stage, you would now think about how to approach the problem. As you will agree, review at this stage too is vital. The selection of the wrong strategy can put the entire system in a miserable state. Reviews performed at this stage will help you in:

- Analysing various strategies to solve the problem;
- Analysing the feasibility of each strategy;
- Knowing the advantages/disadvantages of each strategy.

3. Code review: This involves examining the source code to spot defects. While working on the development of systems software tools, I have been a witness to how code review can really help you find defects. It is good to let someone in your team walk through your code, and do the review. All the members must review the code changes with respect to other modules and give their feedback in case some side-effect is suspected. Many defects like memory leaks, wrong passing of arguments, unreachable code, lack of readability, high complexity and maintainability issues can be identified via the code review. Finding defects at the coding stage, and fixing them immediately, would prove to be less expensive than finding them in the testing stage.

Source code documentation

Maintaining software is a crucial aspect of the software development process. Poorly documented, or rather undocumented code, is quite difficult to maintain. Having somebody modifying your undocumented code can result in disastrous consequences. So, it's important to document your code and the assumptions made by you, while coding.

Keep the documentation close to the source. Multiple